

SECTION BAnswer **all** questions.

9. (a) The properties of elements and their compounds are determined by their structure and bonding.

(i) Diamond and graphite are allotropes of carbon.

- I. Describe the structure and bonding in diamond and in graphite.
Diagrams may be used as part of your answer.

[2]

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- II. State **one** physical property which is common to both diamond and graphite and **one** which is not. Explain **both** properties in terms of the structure and bonding you have described.

[4]

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- (ii) I. Draw a dot and cross diagram to show the formation of the bonding in sodium chloride. [2]

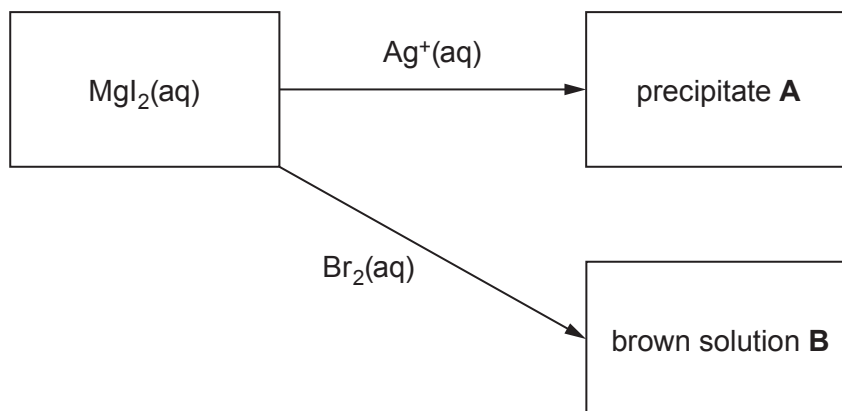
- II. State why the crystal structures of sodium chloride and caesium chloride are different. [1]

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- (b) Two reactions of aqueous magnesium iodide are given in the reaction scheme below.



- (i) State the colour of precipitate **A** and give the **ionic** equation for its formation. [2]

Colour

Ionic equation

- (ii) Name the product that gives solution **B** its brown colour and state the role of aqueous bromine in this reaction. [2]

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- (c) Describe a chemical test, apart from a flame test, which would distinguish between aqueous solutions of magnesium iodide and barium iodide. Your answer should include the reagent(s) used and the observations made in each case. [2]

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